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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are ignored
b. The parts are merged without sorting
c. Each part is sorted independently
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **2**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By breaking the input into smaller parts and merging them
b. By comparing elements sequentially
c. By sorting data in a single pass
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **3**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort does not guarantee sorted order
c. Bubble Sort is less efficient for large lists
d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It is less efficient than custom sorting algorithms
c. It only works with integer arrays
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **6**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. sorted()
c. sort()
d. order()

The correct answer is: sorted()

Question **7**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It does not require recursion
b. It sorts data using a single pass
c. It is less efficient than bubble sort
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **8**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It divides the list into sublists
c. It recursively sorts subproblems
d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **9**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 200,470,80,150,40,90,300,400,70,120
b. 80,150,200,470,40,90,300,400,70,120
c. 40,80,90,150,200,300,400,470,70,120
d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **10**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Complexity
- b. Sorting
- c. Searching
- d. Merging

The correct answer is:
Complexity

Question **11**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Decreases the efficiency of algorithms
- b. Makes data harder to manage
- c. Makes code execution slower
- d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Binary Search
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Inserting data into a list
- c. Arranging data in a particular format
- d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It avoids the need for recursion
- b. It simplifies the sorting process by using only one pass
- c. It allows for efficient parallel processing and sorting of data
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting is rarely used in programming
- b. Sorting helps in finding duplicates quickly
- c. Sorting decreases the efficiency of selection operations
- d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly